Paper D-01: Neural Consciousness: A Constraint-Driven Flux Dynamics (CDFD) Perspective

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Shared Part IV notation and evidence boundaries are defined in `PART_IV_METHODS_AND_CLAIM_BOUNDARY.md`*. This paper states only its local observables, comparison, and failure conditions.*

Abstract

This paper develops a falsifiable CDFD/AFL model of Neural Consciousness. It maps sensory input, recurrent signaling, metabolic support, and task demand to driving flux Φ, energy, attention, connectivity, inhibition, and integration limits to active constraint C, plasticity, arousal, recurrent coordination, and compensatory recruitment to responsiveness S, and learned priors, synaptic state, injury, sleep, and developmental history to structural memory M_s. The target outcome is integration, reportability, performance, and recovery, tested against neurophysiology, imaging, perturbation, behavior, metabolic data, and longitudinal records. The paper treats universality as a shared model architecture rather than an assertion that unlike systems have the same material mechanism. It states the negative result that would make the mapping unnecessary and places the domain claim inside the cumulative CDFD programme and its reproducible runtime boundary.

1 Introduction

A human brain consumes 20% of the body's metabolic energy despite representing only 2% of its mass. This massive energy expenditure is required to maintain the continuous electrical and chemical flux across billions of synaptic junctions. The brain is not a static computer; it is a highly turbulent fluid of information.

To prevent this immense flux from spiraling into chaotic seizures, the brain deploys massive inhibitory constraints. The interplay between excitatory transmission and inhibitory gating creates the highly ordered, complex patterns we experience as cognition.

2 The Cognitive Adaptive Surface

We apply the Adaptive Operating Ratio to neurobiology:

$$\Psi_s = \left(\frac{\Phi}{C}\right) \cdot S \cdot M_s \tag{1}$$

For a specific cortical region:

- **Flux (Φ)**: The rate of excitatory action potentials (e.g., glutamate transmission) and the volume of incoming sensory data.
- **Constraint (C)**: The inhibitory tone of the network (e.g., GABAergic interneurons) and the metabolic limits of the astrocytes providing energy.
- **Responsiveness (S)**: Neuroplasticity—the capacity of the tissue to rapidly grow new dendritic spines or upregulate receptor density to absorb new information.
- **Memory (M_s)**: Long-Term Potentiation (LTP) and structural engrams. The physical reinforcement of a synapse "remembers" past flux, acting as a chronic, semi-permanent modifier to the local topological constraints.

2.1 Epilepsy and PTSD as Topological Failures

A healthy, conscious brain operates exactly at the edge of criticality ($\Psi_s \approx 1$). The flux is maximal, but perfectly contained by the constraints, allowing complex, ordered thought to propagate.

If the inhibitory constraint C drops abruptly (e.g., due to a chemical imbalance or trauma), the excitatory flux Φ runs completely unchecked. The adaptive ratio skyrockets ($\Psi_s \gg 1$). The localized region of the brain is overwhelmed, and the unchecked electrical cascade spreads topologically across the cortex. This is an **Epileptic Seizure**, the neurological equivalent of a supernova or a power grid cascading failure.

Conversely, consider Psychological Trauma (PTSD). A massively terrifying sensory input causes the brain to aggressively enforce a memory lock ($M_s \to 1.0$) on specific amygdala-hippocampal circuits. The brain permanently elevates the constraint C around that specific informational pathway to prevent the traumatic flux from recurring. The local topology becomes rigidly frozen ($\Psi_s \ll 1$). The patient loses cognitive flexibility because the structural memory refuses to adapt to new, safe environments.

3 Measurement Layer

4 Cross-Domain Model Upgrade: Mechanism, Scale, and Tests

4.1 Observable Translation

The paper-specific translation begins with an observable chain rather than a verbal analogy. For Neural Consciousness, the proposed drive is sensory input, recurrent signaling, metabolic support, and task demand. The active limitation is energy, attention, connectivity, inhibition, and integration limits. Responsiveness is carried by plasticity, arousal, recurrent coordination, and compensatory recruitment, while retained state is represented by learned priors, synaptic state, injury, sleep, and developmental history. The outcome to explain is integration, reportability, performance, and recovery.

Table 1: Operational CDFD/AFL map for Paper D-01.

Term	**Paper-specific observable meaning**	**Measurement question**
Φ	sensory input, recurrent signaling, metabolic support, and task demand	What rate, load, gradient, or demand enters the focal system?
C	energy, attention, connectivity, inhibition, and integration limits	Which measured bottleneck limits transfer, function, or recovery?
S	plasticity, arousal, recurrent coordination, and compensatory recruitment	Which process changes effective capacity or reroutes the load?
M_s	learned priors, synaptic state, injury, sleep, and developmental history	Which retained state makes equal present loads produce different futures?
Y	integration, reportability, performance, and recovery	Which outcome is predicted out of sample, with uncertainty?

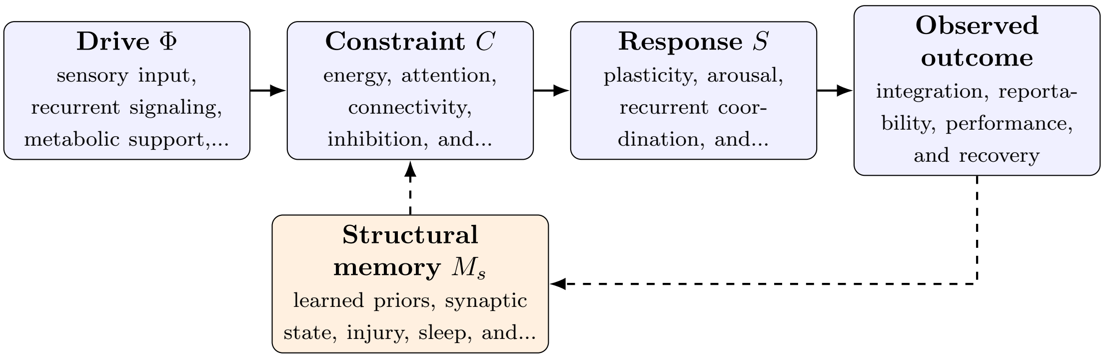

Figure 1: Paper D-01 observable CDFD/AFL architecture for Neural Consciousness. Solid arrows give the proposed forward chain; dashed arrows make history-dependent feedback explicit.

4.2 Paper-Specific Causal Hypothesis

For this paper, the hypothesized causal sequence is specific. A change in sensory input, recurrent signaling, metabolic support, and task demand arrives faster than plasticity, arousal, recurrent coordination, and compensatory recruitment can compensate. This raises or redistributes energy, attention, connectivity, inhibition, and integration limits. If the load persists, the retained state encoded by learned priors, synaptic state, injury, sleep, and developmental history changes the next response, so a later event of equal magnitude can produce a different trajectory in integration, reportability, performance, and recovery. The scientific task is to estimate that sequence against the best ordinary model of the domain, not merely to redescribe the outcome after it occurs.

4.3 Cross-Scale Translation Without Mechanistic Erasure

For the domain applications family, the cross-domain translation is a disciplined test across systems with very different material mechanisms. A valid mapping preserves the baseline science of the domain, declares the scale of every variable, and earns its place through a discriminating prediction rather than resemblance alone. [2, 4, 3, 1] The required scale declaration for this paper is milliseconds to lifetimes; synapses to whole-brain networks. At short scales, the key question is whether the incoming drive is transmitted, buffered, or diverted. At intermediate scales, response changes effective capacity and network exposure. At long scales, retained structure can alter the state space itself. A result is genuinely cross-scale only when the aggregation rule is stated and the same conclusion is not an artifact of averaging.

4.4 Evidence Ladder and Discriminating Tests

Table 2: Evidence ladder for Paper D-01. Each rung can fail independently.

Test	Design	Discriminating result
Proxy audit	Measure neurophysiology, imaging, perturbation, behavior, metabolic data, and longitudinal records and preregister how each record maps to Φ, C, S, M_s, and Y.	The mapping fails if proxies are non-identifiable, circular, or change meaning across cases.
Load ramp	Apply or observe a controlled perturbation of arousal, connectivity, or sensory load while estimating the dominant constraint and response time.	A threshold claim requires a reproducible nonlinear change and uncertainty bounds, not a visually chosen breakpoint.
Recovery and memory	Match present load across units with different histories, then compare recovery paths.	The memory claim requires history-dependent divergence after current conditions are controlled.
Model comparison	Compare the baseline domain model with and without the CDFD/AFL state variables using held-out data.	The paper's stated falsifier is: CDFD variables do not improve discrimination among competing consciousness models.

4.5 Research Programme and Boundary Conditions

The immediate research programme has four steps. First, construct a documented dataset from neurophysiology, imaging, perturbation, behavior, metabolic data, and longitudinal records and retain raw units before normalization. Second, fit a baseline domain model and report its calibration and out-of-sample error. Third, add the smallest identifiable CDFD/AFL state extension and test whether it improves prediction of integration, reportability, performance, and recovery. Fourth, repeat the analysis across the declared scale range and publish negative cases as part of the cross-domain translation audit.

5 Conclusion

Consciousness is the experiential artifact of a system actively regulating its own information flow at the edge of thermodynamic criticality. By understanding the brain as an active CDFD surface, psychiatric and neurological disorders cease to be arbitrary chemical imbalances and instead reveal themselves as highly predictable topological phase failures.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Part-specific runtime diagnostics are under each Part's `outputs/` directory.

Release-wide code: `Part_E_Synthesis/supplementary/`.

Generated summaries: `Part_E_Synthesis/outputs/`.

Figures: `Part_E_Synthesis/figures/`.

6 Reference Layer

References

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